DSA210 Final Report

Alcohol Consumption and Development Indicators

Introduction and Motivation

Alcohol consumption is a major public health and social issue that differs significantly across countries. At the same time, countries also differ in terms of economic development, healthcare quality, education, and overall living standards. Because of these differences, it is interesting to investigate whether alcohol consumption patterns are related to development indicators such as GDP per capita, life expectancy, and Human Development Index (HDI).

The motivation behind this project is to better understand whether socioeconomic development may be associated with alcohol consumption levels. I chose this topic because I wanted to work with real-world international datasets and apply data science techniques to a socially relevant topic. Since the project combines health, economics, and human development indicators, it provided an opportunity to analyze a multidimensional global issue using data-driven methods. Countries with higher income levels and better living standards may have different consumption habits compared to developing countries. Additionally, health-related indicators such as life expectancy may also be connected to alcohol usage patterns.

This project applies data science techniques to analyze international datasets and investigate these relationships using statistical analysis, exploratory data analysis (EDA), visualization, and machine learning methods.

Research Questions

The project focuses on the following research questions:

1. Is alcohol consumption related to GDP per capita?
2. Is alcohol consumption related to life expectancy?
3. Can development indicators help predict alcohol consumption patterns?

Data Sources and Data Collection

The project combines datasets collected from multiple international organizations.

WHO Alcohol Consumption Dataset

The alcohol consumption dataset was collected from the World Health Organization (WHO). The dataset contains alcohol consumption values measured as total alcohol consumption per capita for individuals aged 15 and above.

The dataset includes:

- country information
- year information
- alcohol consumption values
- gender categories
- location type categories

Only country-level observations and "Both sexes" records were used in the analysis.

World Bank GDP per Capita Dataset

GDP per capita data was collected from the World Bank database. GDP per capita is an important economic development indicator because it reflects the average income level of a country.

The original dataset was stored in wide format where each year appeared as a separate column.

World Bank Life Expectancy Dataset

Life expectancy data was also collected from the World Bank database. Life expectancy is commonly used as a measure of health and development.

Similar to the GDP dataset, this dataset was initially stored in wide format.

UNDP HDI Dataset

Human Development Index (HDI) data was collected from the United Nations Development Programme (UNDP). HDI combines indicators related to health, education, and income.

The HDI dataset was used as an additional development indicator for the machine learning section.

Data Preparation and Cleaning

The datasets required several preprocessing and cleaning steps before analysis.

Alcohol Dataset Cleaning

The alcohol dataset was filtered to:

- keep only country-level observations
- keep only “Both sexes” data
- remove unnecessary columns
- convert year and alcohol values into numeric format
- remove missing values
- limit observations to the 2010–2020 period

Columns were renamed for consistency:

- Location → Country
- Period → Year
- FactValueNumeric → Alcohol

GDP and Life Expectancy Dataset Cleaning

The GDP and life expectancy datasets were originally stored in wide format. Therefore, they were transformed into long format using pandas melt operations.

The cleaning process included:

- converting years into numeric format
- removing missing values
- filtering years between 2010 and 2020
- removing aggregate groups such as “World” and “High income”

Country Name Standardization

Some country names were inconsistent across datasets. Therefore, country names were standardized before merging.

Examples include:

- Türkiye → Turkey
- United States of America → United States
- Viet Nam → Vietnam

Dataset Merging

After cleaning, the datasets were merged using country and year information.

The final merged dataset included:

- Country
- Year
- Alcohol consumption
- GDP per capita
- Life expectancy

The cleaned merged dataset was exported as:

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panel_dataset.csv
```

An extended dataset including HDI values was also created.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure of the data and identify potential relationships between variables.

Dataset Overview

The notebook first examined:

- the first rows of the dataset
- dataset dimensions
- variable types
- missing values
- descriptive statistics

This step helped verify that the datasets were successfully cleaned and merged.

Distribution Analysis

Histograms and visualizations were used to examine:

- alcohol consumption distribution
- GDP per capita distribution
- life expectancy distribution

The analysis showed that GDP per capita values were highly skewed because some countries have significantly larger economies.

Correlation Analysis

Correlation heatmaps and scatter plots were used to visually examine relationships between variables.

The visualizations suggested:

- a positive relationship between alcohol consumption and GDP per capita
- a relationship between alcohol consumption and life expectancy

These findings motivated the hypothesis testing section.

Hypothesis Testing

The project used Pearson correlation tests to statistically evaluate the relationships between variables.

Hypothesis Test 1

Relationship Between Alcohol Consumption and GDP per Capita

Null Hypothesis (H0):
There is no relationship between alcohol consumption and GDP per capita.

Alternative Hypothesis (H1):
There is a relationship between alcohol consumption and GDP per capita.

The Pearson correlation test produced:

- a positive correlation coefficient
- a p-value below 0.05

Since the p-value was smaller than 0.05, the null hypothesis was rejected.

This result suggests that alcohol consumption and GDP per capita are statistically related.

Hypothesis Test 2

Relationship Between Alcohol Consumption and Life Expectancy

Null Hypothesis (H0):
There is no relationship between alcohol consumption and life expectancy.

Alternative Hypothesis (H1):
There is a relationship between alcohol consumption and life expectancy.

The Pearson correlation test again produced a statistically significant result with a p-value below 0.05.

Therefore, the null hypothesis was rejected.

The analysis suggests that alcohol consumption and life expectancy are statistically related.

Machine Learning Section

Machine learning methods were applied to further analyze alcohol consumption patterns.

The machine learning workflow included:

- feature selection
- train-test split
- model training
- model evaluation
- interpretation of model performance

Development indicators such as GDP per capita, life expectancy, and HDI were used as predictors.

The goal of the machine learning section was not only prediction but also understanding whether development indicators provide meaningful information about alcohol consumption patterns.

The results showed that development-related variables contributed useful information for predicting alcohol consumption levels.

Findings

The project produced several important findings.

1. Alcohol consumption showed a statistically significant positive relationship with GDP per capita.
2. Alcohol consumption also showed a statistically significant relationship with life expectancy.
3. Countries with higher development indicators often displayed different alcohol consumption patterns compared to less developed countries.
4. Machine learning models demonstrated that socioeconomic indicators contain meaningful predictive information.
5. The project showed that international datasets from different organizations can be successfully cleaned, merged, and analyzed together.

Limitations and Future Work

Although the project produced meaningful findings, several limitations exist.

Limitations

- Alcohol consumption data is based on reported estimates.
- HDI values were not available for all years.
- Correlation does not imply causation.
- Some countries had incomplete observations.
- Cultural and legal differences between countries were not directly modeled.

Future Work

Future studies could:

- include additional health indicators
- analyze regional differences separately
- use time-series forecasting methods
- apply more advanced machine learning models
- investigate causal relationships instead of only correlations

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Conclusion

This project investigated the relationship between alcohol consumption and development indicators using international datasets from WHO, World Bank, and UNDP.

The analysis included data cleaning, exploratory data analysis, visualization, hypothesis testing, and machine learning methods.

The results showed statistically significant relationships between alcohol consumption and important development indicators such as GDP per capita and life expectancy.

Overall, the project demonstrates how data science techniques can be used to combine multiple real-world datasets and generate meaningful insights about global socioeconomic patterns.